

Lab08

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a biopsy data set describe characteristics of the cell nuclei present in digitized images of a fine needle aspiration (FNA) of a breast mass.

##DATA IMPORT

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names =1)
```

```
head(wisc.df,3)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.8	1001
842517	M	20.57	17.77	132.9	1326
84300903	M	19.69	21.25	130.0	1203
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001		0.14710
842517	0.08474	0.07864	0.0869		0.07017
84300903	0.10960	0.15990	0.1974		0.12790
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se

842302	0.2419		0.07871	1.0950	0.9053	8.589
842517	0.1812		0.05667	0.5435	0.7339	3.398
84300903	0.2069		0.05999	0.7456	0.7869	4.585
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se	
842302	153.40	0.006399	0.04904	0.05373		0.01587
842517	74.08	0.005225	0.01308	0.01860		0.01340
84300903	94.03	0.006150	0.04006	0.03832		0.02058
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst		
842302	0.03003		0.006193	25.38		17.33
842517	0.01389		0.003532	24.99		23.41
84300903	0.02250		0.004571	23.57		25.53
	perimeter_worst	area_worst	smoothness_worst	compactness_worst		
842302	184.6	2019	0.1622			0.6656
842517	158.8	1956	0.1238			0.1866
84300903	152.5	1709	0.1444			0.4245
	concavity_worst	concave.points_worst	symmetry_worst			
842302	0.7119		0.2654			0.4601
842517	0.2416		0.1860			0.2750
84300903	0.4504		0.2430			0.3613
	fractal_dimension_worst					
842302		0.11890				
842517		0.08902				
84300903		0.08758				

Make sure we remove or exclude the diagnosis column from the data-set that we use for further analysis

```
wisc.data <- wisc.df[,-1]
diagnosis <- as.factor(wisc.df$diagnosis)
```

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

569 observations

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B    M  
357 212
```

212 are malignant

Q3. How many variables/features in the data are suffixed with `_mean`?

```
colnames(wisc.data)
```

```
[1] "radius_mean"      "texture_mean"  
[3] "perimeter_mean"   "area_mean"  
[5] "smoothness_mean"  "compactness_mean"  
[7] "concavity_mean"    "concave.points_mean"  
[9] "symmetry_mean"     "fractal_dimension_mean"  
[11] "radius_se"         "texture_se"  
[13] "perimeter_se"      "area_se"  
[15] "smoothness_se"     "compactness_se"  
[17] "concavity_se"      "concave.points_se"  
[19] "symmetry_se"       "fractal_dimension_se"  
[21] "radius_worst"      "texture_worst"  
[23] "perimeter_worst"   "area_worst"  
[25] "smoothness_worst"  "compactness_worst"  
[27] "concavity_worst"   "concave.points_worst"  
[29] "symmetry_worst"    "fractal_dimension_worst"
```

```
length( grep("_mean", colnames(wisc.data), value=T))
```

```
[1] 10
```

10 are suffixed with mean.

Principal Component Analysis

We need to scale our data before PCA with the `scale=TRUE` ARGUMENT TO `PRCOMP()`.

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

```
colMeans(wisc.data)
```

radius_mean	texture_mean	perimeter_mean
1.412729e+01	1.928965e+01	9.196903e+01
area_mean	smoothness_mean	compactness_mean
6.548891e+02	9.636028e-02	1.043410e-01
concavity_mean	concave.points_mean	symmetry_mean
8.879932e-02	4.891915e-02	1.811619e-01
fractal_dimension_mean	radius_se	texture_se
6.279761e-02	4.051721e-01	1.216853e+00
perimeter_se	area_se	smoothness_se
2.866059e+00	4.033708e+01	7.040979e-03
compactness_se	concavity_se	concave.points_se
2.547814e-02	3.189372e-02	1.179614e-02
symmetry_se	fractal_dimension_se	radius_worst
2.054230e-02	3.794904e-03	1.626919e+01

texture_worst	perimeter_worst	area_worst
2.567722e+01	1.072612e+02	8.805831e+02
smoothness_worst	compactness_worst	concavity_worst
1.323686e-01	2.542650e-01	2.721885e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
1.146062e-01	2.900756e-01	8.394582e-02

```
apply(wisc.data,2,sd)
```

radius_mean	texture_mean	perimeter_mean
3.524049e+00	4.301036e+00	2.429898e+01
area_mean	smoothness_mean	compactness_mean
3.519141e+02	1.406413e-02	5.281276e-02
concavity_mean	concave.points_mean	symmetry_mean
7.971981e-02	3.880284e-02	2.741428e-02
fractal_dimension_mean	radius_se	texture_se
7.060363e-03	2.773127e-01	5.516484e-01
perimeter_se	area_se	smoothness_se
2.021855e+00	4.549101e+01	3.002518e-03
compactness_se	concavity_se	concave.points_se
1.790818e-02	3.018606e-02	6.170285e-03
symmetry_se	fractal_dimension_se	radius_worst
8.266372e-03	2.646071e-03	4.833242e+00
texture_worst	perimeter_worst	area_worst
6.146258e+00	3.360254e+01	5.693570e+02
smoothness_worst	compactness_worst	concavity_worst
2.283243e-02	1.573365e-01	2.086243e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
6.573234e-02	6.186747e-02	1.806127e-02

```
# Perform PCA on wisc.data by completing the following code
```

```
wisc.pr <- prcomp( wisc.data, scale= TRUE)
```

```
# Look at summary of results
```

```
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
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Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
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Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
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Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

PC1 captures about 44.27% of the original variance.

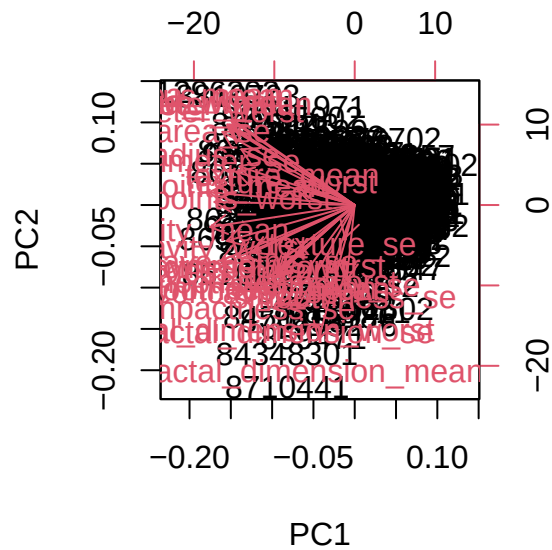
Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

It takes 3 principal components are required to describe 70% of the variance.

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

It takes 7 principal components to describe 90% of the variance.

```
biplot(wisc.pr)
```

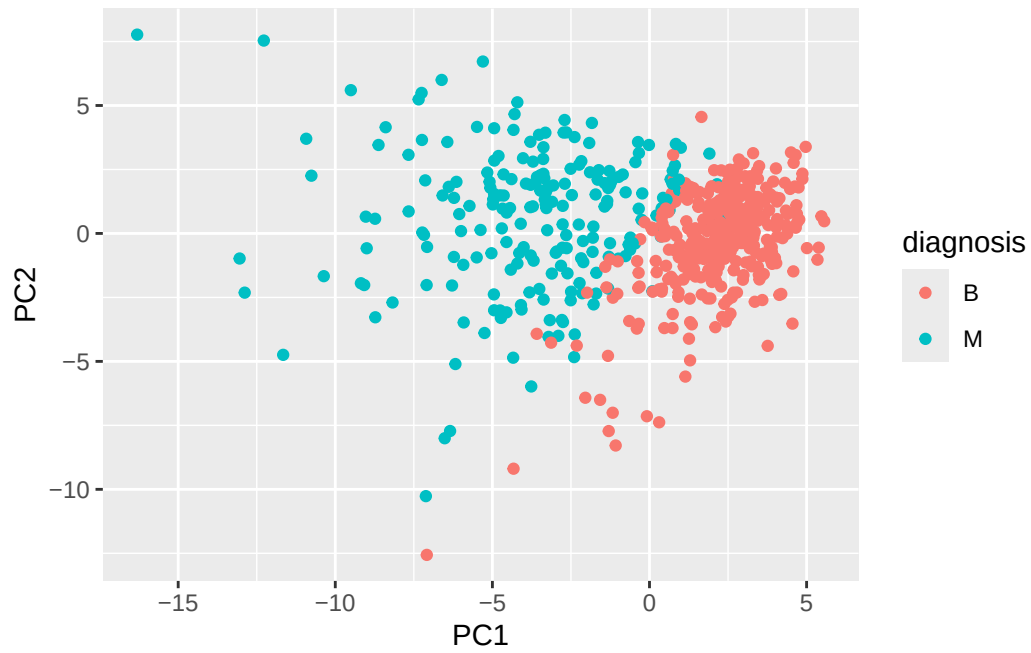


Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

It is very difficult to understand the PCA model as there is many overlap and clustering due to many variables being plotted at once. It is not clear on what the graph is depicting and identify patterns.

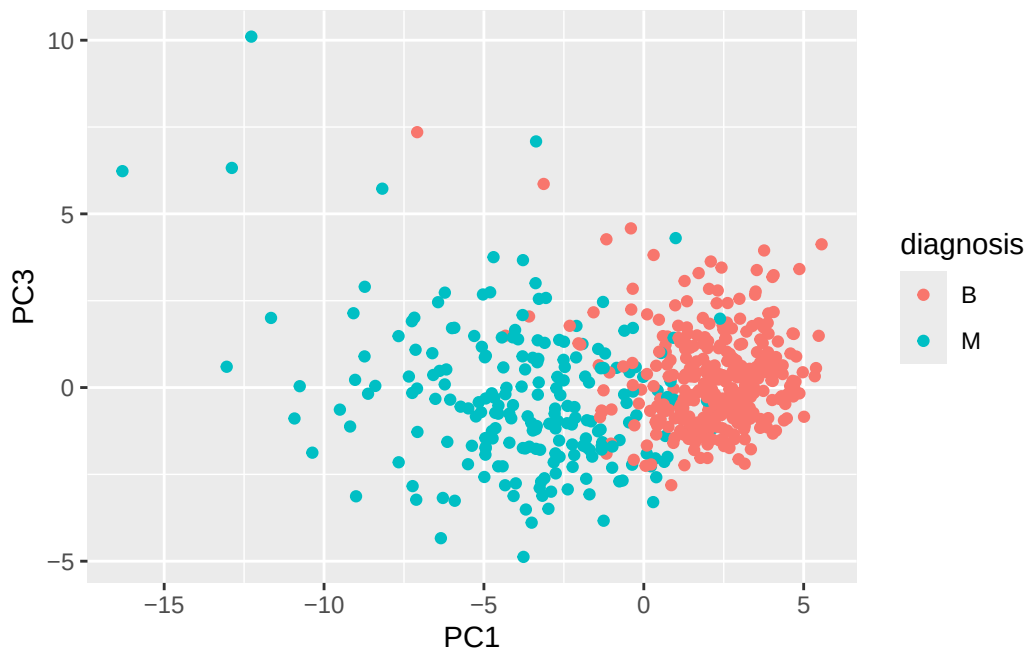
```
library(ggplot2)

ggplot(wisc.pr$x)+
  aes(PC1,PC2, col=diagnosis)+
  geom_point()
```



```
library(ggplot2)

ggplot(wisc.pr$x)+
  aes(PC1,PC3, col=diagnosis)+
  geom_point()
```

Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

I notice that the points are dependent on PC1 and that is what causes the division between malignant and benign. The separation is on the PC1 axis.

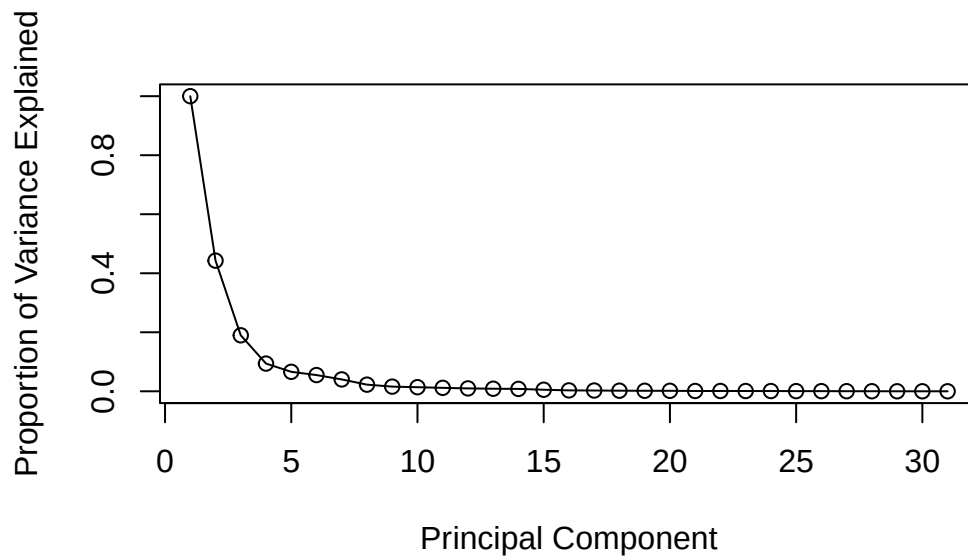
```
pr.var <- (wisc.pr$sdev)^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

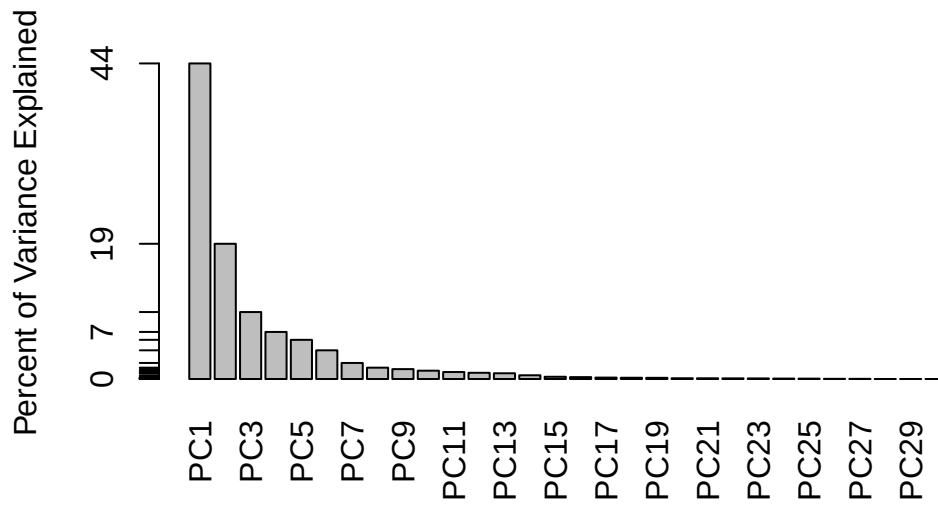
```
pve <- pr.var / sum(pr.var)
pve
```

```
[1] 4.427203e-01 1.897118e-01 9.393163e-02 6.602135e-02 5.495768e-02
[6] 4.024522e-02 2.250734e-02 1.588724e-02 1.389649e-02 1.168978e-02
[11] 9.797190e-03 8.705379e-03 8.045250e-03 5.233657e-03 3.137832e-03
[16] 2.662093e-03 1.979968e-03 1.753959e-03 1.649253e-03 1.038647e-03
[21] 9.990965e-04 9.146468e-04 8.113613e-04 6.018336e-04 5.160424e-04
[26] 2.725880e-04 2.300155e-04 5.297793e-05 2.496010e-05 4.434827e-06
```

```
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



```
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```
sort(abs(wisc.pr$rotation[, 1]), decreasing = TRUE)
```

concave.points_mean	concavity_mean	concave.points_worst
0.26085376	0.25840048	0.25088597
compactness_mean	perimeter_worst	concavity_worst
0.23928535	0.23663968	0.22876753
radius_worst	perimeter_mean	area_worst
0.22799663	0.22753729	0.22487053
area_mean	radius_mean	perimeter_se
0.22099499	0.21890244	0.21132592
compactness_worst	radius_se	area_se
0.21009588	0.20597878	0.20286964
concave.points_se	compactness_se	concavity_se
0.18341740	0.17039345	0.15358979
smoothness_mean	symmetry_mean	fractal_dimension_worst
0.14258969	0.13816696	0.13178394
smoothness_worst	symmetry_worst	texture_worst

0.12795256	0.12290456	0.10446933
texture_mean	fractal_dimension_se	fractal_dimension_mean
0.10372458	0.10256832	0.06436335
symmetry_se	texture_se	smoothness_se
0.04249842	0.01742803	0.01453145

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

the loading value is -0.2608538. This is one of the largest contributor but other features such as `concavity_mean` and `concave.points_worst` have similar and large contributions.

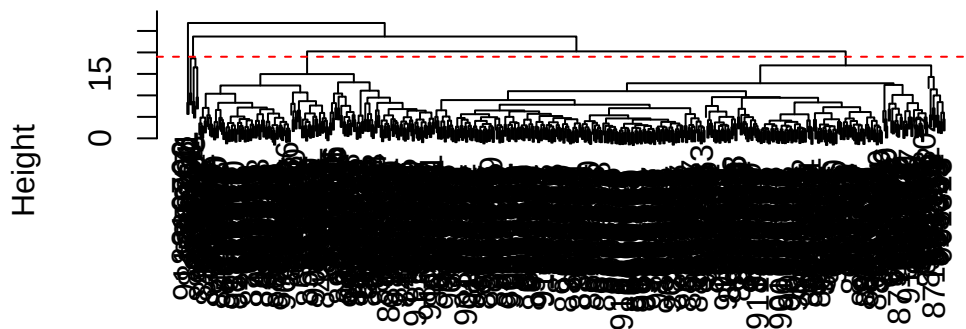
Hierachical Clustering

Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

The height for cutting the model into 4 clusters is about 19.

```
wisc.hclust <- hclust( dist( scale(wisc.data) ) )
plot(wisc.hclust)
abline(h=19,col="red", lty =2)
```

Cluster Dendrogram



```
dist(scale(wisc.data))
hclust (*, "complete")
```

```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)

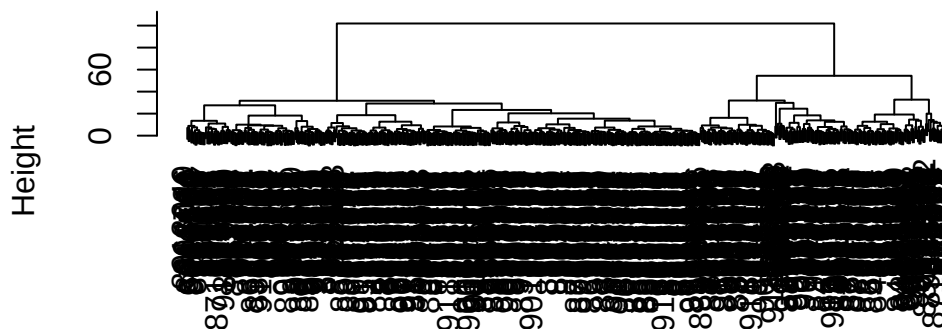
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

```
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
```

```
wisc.hclust.ward <- hclust(data.dist, method = "ward.D2")
plot(wisc.hclust.ward)
```

Cluster Dendrogram



```
data.dist
hclust (*, "ward.D2")
```

```
table(cutree(wisc.hclust.ward, k = 4), diagnosis)
```

diagnosis

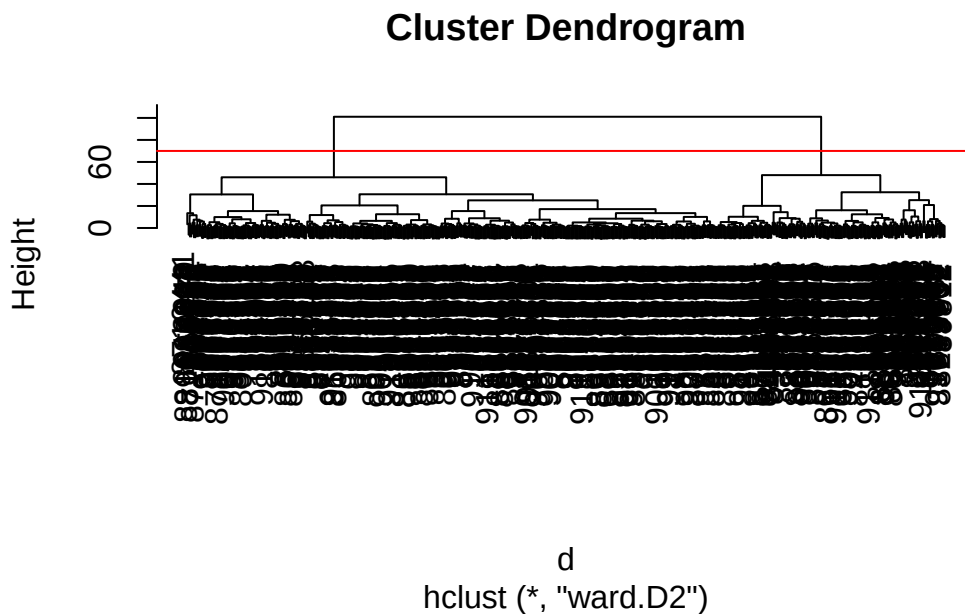
	B	M
1	0	115
2	6	48
3	337	48
4	14	1

Q12. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

I prefer ward.D2 as it provides a more clear separation between the two samples as compared to the linkage method. By minimizing the variation it makes the clustering more clear and reliable.

Combining Methods

```
d <- dist(wisc.pr$x[,1:4])
wisc.pr.hclust <- hclust(d, method="ward.D2")
plot(wisc.pr.hclust)
abline(h=70, col="red")
```



```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1  2
171 398
```

How does this clustering `grps` correspond to the expert `diagnosis`?

```
table(diagnosis)
```

```
diagnosis
  B  M
357 212
```

```
table(diagnosis, grps)
```

```
      grps
diagnosis  1  2
      B   6 351
      M 165  47
```

```
table(wisc.hclust.clusters, diagnosis)
```

```
      diagnosis
wisc.hclust.clusters  B  M
      1  12 165
      2   2   5
      3 343  40
      4   0   2
```

Q13. How well does the newly created `hclust` model with two clusters separate out the two “M” and “B” diagnoses?

The PCA-based `hclust` separates the diagnosis fairly ok. While one cluster has 165 malignant and 6 benign the other cluster has 47 malignant and 351 benign.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses

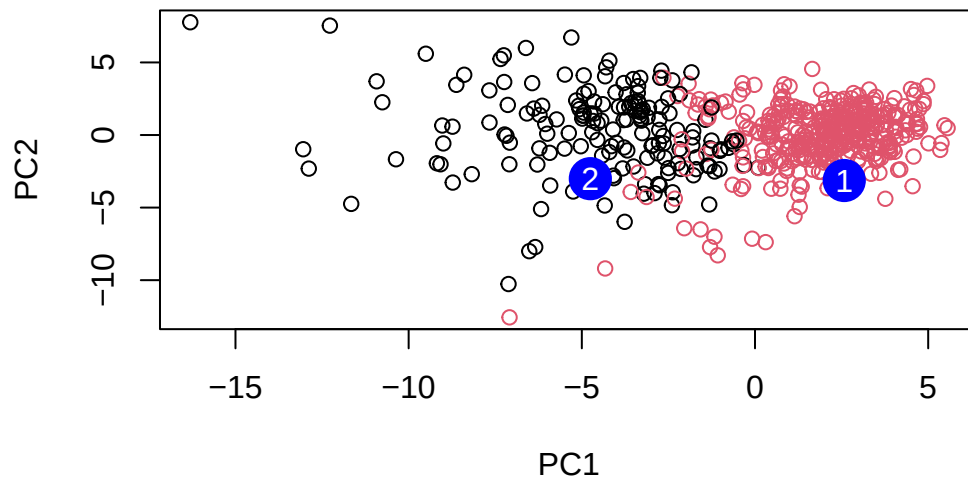
The clustering without PCA is not as efficient in separating the diagnoses, for example, malignant and benign are mixed across multiple clusters and this creates an overlap. PCA clustering separates data more clearly into two separate groups with group 1 containing 165 m and 6 b while group 2 has 351 b and 47 m. With PCA it captures the difference between the two groups more clearly it is still not a perfect way to separate them.

Prediction

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```

Q16. Which of these new patients should we prioritize for follow up based on your results?

Patient 2 should be prioritized for a follow up because it is closer to the malignant cluster on the plot and patient 1 is closer to the benign area.